Revision Summary (October 10, 2026)

As specified, only the mother drone can run the autonomy software. The worker parts list has no barometer and no compass, so the workers cannot fly GPS waypoints. The baseline demonstration is therefore the real ArduPilot mother drone plus simulated agents under one coordinator, pending the faculty advisor's written approval. Six decisions are needed at the October 16 design review; they are listed at the end of this summary.

This revision keeps the original plan text and marks every change. **[NEW]** marks added content. **[REVISED]** marks changed content; where wording was replaced, the original follows as *(Was: ...)*. Text without a marker is unchanged from the original plan.

Main changes

1. **Autonomy baseline:** the real mother plus two simulated agents, through the same MAVLink interface, is the baseline demonstration (Sections 1.1 and 7). The workers remain physical prototypes for stabilized flight, sensing, camera streaming, and telemetry.
2. **Communications:** every vehicle talks directly to the ground station, and the mother no longer relays traffic (moved to future work). MAVLink v2 is the common message format. DroneBridge for ESP32 carries the mother's MAVLink over ESP-NOW, a USB ESP32 gateway connects the laptop to the workers' ESP-NOW, and video uses plain Wi-Fi to a team-owned access point (Section 5.3).
3. **Local LLM:** the deterministic planner makes every mission decision. The LLM turns operator briefs into validated JSON mission specs and writes after-action summaries, and the system runs the same with it switched off (Section 5.5).
4. **Vision:** search target, altitude, dataset, evaluation size, and geotagging are now defined requirements (Section 5.4).
5. **Swarm coordinator:** vehicle-adapter interface, coverage grid, capability-weighted allocation, and reassignment of unsearched cells (Section 5.6).
6. **Mother flight controller:** buy the DolphinRC F405 V3 only if an ArduPilot target with matching gyro and barometer chips is confirmed. Otherwise use the SpeedyBee F405 V3 50A stack, which has documented ArduPilot support and a built-in barometer.
7. **Flight authorization:** UCI EH&S approval through the UC Drone Web App, TRUST certificates, FAA registration, and Remote ID for the mother. Moved from Week 7 to Week 3.
8. **Budget:** Section C itemized, spare propellers and gateway hardware added, sales tax shown separately, and Worker 2's funds held in reserve until the October 30 gate (Section 4.1).
9. **Performance targets:** rewritten as measurable tests (Section 6.4).
10. **Workers:** evaluate Espressif's ESP-Drone firmware before writing a stabilizer from scratch.
11. **Team roster:** the header lists two Computer Engineering students, but the roles list one. The second Computer Engineering student should lead the worker drone.
12. **Risks:** six new risks added to the register (Section 7).

Decisions needed at the October 16 design review

1. Advisor: does the baseline demonstration (real mother plus simulated agents) satisfy the capstone? Get the answer in writing.
2. Flight controller: DolphinRC (only if its ArduPilot target is confirmed) or SpeedyBee F405 V3.
3. RC link and Remote ID: borrow, buy, or fly the mother only at an FAA-Recognized Identification Area.
4. Search target, search-area size, and flight altitude.
5. Roster: who leads the worker drone.
6. Test site approved by UCI EH&S.

A. Mother drone — 1 aircraft

Target: 5-inch, 3S, brushless, ArduPilot, GPS/compass, ESP32 video

Item and specification	Qty	Where to buy	Unit price	Total
[REVISED] SpeedyBee F405 V3 50A FC + ESC stack (documented ArduPilot support, built-in barometer; price checked October 2026). Buy the DolphinRC F405 V3 50A stack ($54.59, SoarSky) instead only if an ArduPilot target with matching gyro and barometer chips is confirmed by October 16. *(Was: DolphinRC F405 V3 50A FC + ESC stack, SoarSky, $54.59.)*	1	RaceDayQuads	$69.99	$69.99
RS2205 2300KV brushless motors, 4-piece variant	1 pack	B-CUBE RC	$20.09	$20.09
5.1-inch Gemfan propellers, 2 CW + 2 CCW. **[REVISED]** 1 set plus 2 spare sets. *(Was: 1 set.)*	3 sets	Pyrodrone — 51499	$3.99	$11.97
BN-880 GPS with integrated compass	1	Exo Express	$12.00	$12.00
AI-Thinker ESP32-CAM + OV2640. **[REVISED]** Buy in Week 3 for vision dataset collection, then mount facing straight down.	1	ElectroPeak	$6.95	$6.95
Powerhobby 3S 1500mAh 45C XT60 LiPo	1	WREKD	$23.99	$23.99
[NEW] Second Powerhobby 3S 1500mAh LiPo. Borrow first; buy only if the Section 4.1 remainder allows.	1	WREKD or loan	$23.99	Unfunded

5-inch printed frame, GPS mast and camera mount	1	UCI fabrication	$0	$0
[NEW] ESP32 development board running DroneBridge for ESP32 (air unit), wired to a spare flight-controller UART for MAVLink telemetry	1	ElectroPeak	$4.95	$4.95
Mother total. **[REVISED]** *(Was: $121.61.)*				$149.94

[NEW] Estimated all-up weight is about 400 g. Under the recreational rules, any aircraft over 250 g must be FAA-registered and broadcast Remote ID (see Section C).

B. Worker drones — 2 identical aircraft

Each worker: custom ESP32 flight control, 8520 brushed motors, IMU, camera, GNSS, 1S battery

Item and specification	**Qty for 2 drones**	**Where to buy**	**Unit price**	**Total**
ESP32-WROOM-32D development board. **[REVISED]** If ESP-Drone is adopted, confirm it builds for this chip; Espressif's documentation lists the ESP32-S2 and ESP32-S3 as its maintained targets.	2	ElectroPeak	$4.95	$9.90
GY-521 MPU6050 IMU	2	ElectroPeak	$1.96	$3.92
8520 brushed motors, 4 motors + propellers per kit	2 kits	SpeedyFPV	$4.49	$8.98
AI-Thinker ESP32-CAM OV2640	2	ElectroPeak	$6.95	$13.90
ATGM336H GPS with ceramic antenna	2	Muzi Works	$3.46	$6.92
1S 650mAh 60C PH2.0 LiHV battery	2	WREKD	$8.49	$16.98
MT3608 adjustable 5V boost module	2	ElectroPeak	$0.33	$0.66
Four-channel MOSFET motor-driver components, target allocation	2 sets	DigiKey — SI2302 + UCI prototyping supplies	$2.50	$5.00
Printed frames, prop guards, brackets	2	UCI fabrication	$0	$0
Worker total (both). **[REVISED]** Worker 2's half ($33.13) is held in reserve until the October 30 propulsion gate.				$66.26

[NEW] **Sensing limits:** this list has no barometer and no magnetometer, so the workers cannot

hold altitude or heading and cannot fly GPS waypoints. ESP-Drone offers height-hold and position-hold only with extension boards, which are not budgeted.

[NEW] Mass and thrust estimate: about 90–110 g all-up against about 120–180 g of total thrust from four 8520 motors on 1S, a thrust-to-weight ratio of about 1.2–1.8. The target is at least 2:1. The October 30 gate decides; plan as if it fails.

[NEW] Firmware base: evaluate Espressif's ESP-Drone before writing a stabilizer. It is open-source firmware ported from Crazyflie, and its reference board pairs an ESP32-S2 with an MPU6050, the same IMU used here. Espressif has offered only limited support since December 2022.

C. Radio, programming, shared hardware, and compliance [NEW]

This section itemizes the original "Radio, programming, and shared hardware" budget line ($70.86), which was not broken down. Prices marked "est." are estimates to verify before purchase.

Item and specification	Qty	Where to buy	Unit price	Total
ESP32 development board running DroneBridge for ESP32 (ground unit), on the laptop's USB port	1	ElectroPeak	$4.95	$4.95
ESP32 development board as the worker ESP-NOW to USB gateway (a laptop's Wi-Fi cannot send or receive ESP-NOW)	1	ElectroPeak	$4.95	$4.95
ESP32-CAM USB programmer (ESP32-CAM-MB or USB-UART adapter); borrow if UCI has one	1	TBD	est. $4.00	$4.00
FAA recreational registration (one per registrant; covers that person's recreational aircraft)	1	FAA DroneZone	$5.00	$5.00
TRUST certificate for each pilot	1 per pilot	FAA online test	$0	$0
Wi-Fi access point for video (laptop hotspot or borrowed travel router; not campus Wi-Fi)	1	Loan	$0	$0
RC transmitter and receiver for the mother's safety link (original plan item, never itemized)	1	Loan first	est. $50–70 if bought	From remainder
Remote ID broadcast module for the mother (required over 250 g unless flown only at an FAA-Recognized Identification Area)	1	TBD	est. $30–50	From remainder
Section C itemized total				$18.90

UCI SENIOR DESIGN PROJECT

Autonomous Search-and-Rescue Drone Swarm

Fall 2026 Project Execution Plan | EECS 159A

University: University of California, Irvine
Project Duration: Fall 2026 – Winter 2027
Current Date: October 10, 2026 (end of Fall Week 2) **[REVISED]** *(Was: October 7, 2026 (Fall Week 2).)*
Fall Quarter End: December 11, 2026
Final Project Deadline: March 2027
Team Size: 4 students (2 Computer Engineering, 2 Computer Science & Engineering)
[REVISED] Section 2 assigns 1 Computer Engineering and 3 Computer Science & Engineering roles; confirm the roster.
Hardware Budget: $300 maximum
Software Budget: $0 (open-source software)

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1. PROJECT OVERVIEW

1.1 Project Description

The objective of this senior design project is to develop a low-cost, autonomous search-and-rescue drone swarm consisting of three quadcopters and a ground-based artificial intelligence system.

The system will consist of:

- **1 Mother Drone:** A larger quadcopter responsible for GPS navigation and autonomous waypoint missions. **[REVISED]** The communication relay role is removed from scope and moved to future work; every vehicle talks directly to the ground station. *(Was: "...GPS navigation, autonomous waypoint missions, and acting as a communication relay.")*
- **2 Worker Drones:** Smaller quadcopters designed to explore assigned regions, capture images, and transmit sensor information. **[REVISED]** As specified (no barometer or compass), the workers cannot fly GPS waypoints. They are built as physical prototypes for stabilized flight, sensing, camera streaming, and telemetry. In the autonomy demonstration, their search role is flown by simulated agents unless altitude and heading sensing is added.
- **1 Ground Control Station:** A laptop responsible for artificial intelligence, object detection, mission planning, telemetry monitoring, and swarm coordination.

The system will use ArduPilot, ESP32 microcontrollers, ESP-NOW/Wi-Fi communications,

YOLOv8, and a locally hosted large language model. **[REVISED]** MAVLink v2 is the common message format for every vehicle and for the simulator.

The primary goal is to demonstrate how multiple inexpensive drones can collaborate to locate simulated victims or objects of interest within a designated search region.

[NEW] Baseline demonstration: the real ArduPilot mother drone and two simulated agents run together under one coordinator, through the same MAVLink interface, with live detection on the mother's video. The faculty advisor must confirm this baseline in writing.

1.2 Project Constraints

1. Maximum hardware budget of $300, including taxes and shipping.
2. All software must be free and open source.
3. All major physical components should be sourced by the end of October where feasible.
4. UCI facilities will be used for fabrication and electronics prototyping.
5. The project must be completed by the end of Winter Quarter 2027.
6. Fall Quarter will focus on design, procurement, early hardware development, and preliminary integration.
7. Winter Quarter will focus on system integration, autonomous behavior, testing, and final demonstrations.
8. All physical flight tests must comply with applicable safety, UCI, and FAA requirements. **[REVISED]** These include: prior approval and post-flight reporting through the UC Drone Web App (UCI EH&S); Class C controlled airspace over campus, with additional FAA and John Wayne Airport restrictions; no flights over people or near buildings; a TRUST certificate for each pilot; FAA registration and Remote ID for any aircraft over 250 g; and no simultaneous multi-drone flight without written EH&S approval (under Part 107, one person flying several drones needs a §107.35 waiver).

1.3 Fall Quarter Objectives

By December 11, 2026, the team should aim to have:

- Finalized and approved system architecture.
- Complete bill of materials and procurement records.
- Mother drone assembled and configured with ArduPilot.
- Mother drone successfully bench-tested and, if authorized, flight-tested.
- At least one functional worker drone prototype.
- Worker drone thrust, mass, power, and flight-control feasibility evaluated.
- ESP32 Wi-Fi and ESP-NOW communications demonstrated. **[REVISED]** ESP-NOW carries MAVLink telemetry; Wi-Fi carries video.
- Camera stream successfully transmitted to the ground station.
- YOLOv8 object detection running on the ground laptop.
- Three-agent autonomous search simulation running. **[REVISED]** Running through the same MAVLink adapter the real mother uses.
- Preliminary integration between hardware and ground software.
- Complete documentation, CAD files, wiring diagrams, and test results.
- A detailed Winter Quarter development plan.

- **[NEW]** Flight authorization in place before any outdoor flight: UC Drone Web App approval, TRUST certificates, FAA registration, and a Remote ID plan.
- **[NEW]** Search target, search area, and flight altitude defined; detector evaluated on at least 200 held-out target instances.
- **[NEW]** Faculty advisor's written decision on the baseline demonstration.

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2. TEAM RESPONSIBILITIES

Team Member 1 — Computer Engineering: Mother Drone Lead

Primary Responsibilities:

- Research and select the ArduPilot-compatible flight controller. **[REVISED]** Before purchase, confirm an ArduPilot firmware target exists for the exact board, with matching gyro and barometer chips.
- Select compatible motors, ESCs, propellers, and batteries.
- Design and assemble the mother drone.
- Configure ArduPilot firmware and Mission Planner.
- Install and configure GPS and compass.
- Integrate the physical safety-control receiver.
- Calibrate sensors, motors, and flight modes.
- Develop and execute flight-readiness procedures.
- Perform authorized flight testing.
- Document the mother drone hardware and test results.
- **[NEW]** Install the DroneBridge telemetry ESP32 on a spare flight-controller UART.
- **[NEW]** Configure ArduPilot's polygon geofence and ground-station-loss failsafe.
- **[NEW]** Mount the ESP32-CAM facing straight down and record its lens field of view.

Fall Deliverable: Assembled, configured, and tested mother drone.

Team Member 2 — Computer Science & Engineering: Worker Drone Lead

[REVISED] The header lists two Computer Engineering students, but only Team Member 1 is assigned as one. Recommended: the second Computer Engineering student leads this role, which carries the most hardware work (motor drivers, power electronics, PCB).

Primary Responsibilities:

- Research micro-drone propulsion and flight-controller options.
- Design worker drone frames and mounting systems.
- Develop the worker hardware mass and power budgets.
- Integrate ESP32, IMU, camera, GPS, and motor drivers.

- Develop embedded sensor and motor-control firmware. **[REVISED]** Evaluate Espressif's ESP-Drone firmware as the base before writing a custom stabilizer.
- Implement stabilization and emergency disarm logic.
- Test motors, batteries, regulators, and communications.
- Evaluate custom PCB feasibility.
- Construct and validate at least one worker prototype.
- **[NEW]** Send MAVLink HEARTBEAT, GLOBAL_POSITION_INT, and SYS_STATUS messages over ESP-NOW.
- **[NEW]** Document the workers' sensing limits (no barometer or compass) and which flight modes they allow.

Fall Deliverable: Functional worker prototype with documented flight feasibility.

Team Member 3 — Computer Science & Engineering: AI and Ground Station Lead

Primary Responsibilities:

- Configure the ground laptop environment. **[REVISED]** Record its GPU, RAM, and OS; YOLO on up to three streams plus a local LLM is tight without a GPU.
- Install and configure YOLOv8.
- Develop the search-and-rescue object detection pipeline. **[REVISED]** Fine-tune on aerial data (VisDrone, HERIDAL, SARD) plus the team's overhead images (Section 5.4).
- Receive live ESP32 camera streams.
- Develop a ground station dashboard.
- Display drone telemetry and connection status.
- Record detections, timestamps, and confidence scores.
- Integrate camera detection with mission-state messages.
- Measure inference speed, video latency, and detection accuracy.
- **[NEW]** Define the search target with the team by October 16 and build the overhead image dataset with the actual ESP32-CAM.
- **[NEW]** Geotag detections using camera orientation, field of view, altitude, attitude, and synchronized timestamps.
- **[NEW]** Measure video stream quality at 10, 25, and 50 m in Week 4.

Fall Deliverable: Operational AI detection and ground monitoring software.

Team Member 4 — Computer Science & Engineering: Swarm Autonomy Lead

Primary Responsibilities:

- Develop the swarm mission coordinator. **[REVISED]** Build it on a vehicle-adapter interface with three implementations: MAVLink (SITL and the real mother), worker, and virtual agent (Section 5.6).
- Set up ArduPilot software-in-the-loop (SITL) simulation.
- Implement search-area division and task assignment. **[REVISED]** Coverage grid of about 5 m cells, with area split by each vehicle's speed × endurance.

- **[REVISED]** Adopt MAVLink v2 as the system-wide protocol, and define each vehicle's message set and system ID. *(Was: Design the telemetry and communication protocol.)*
- Integrate ESP-NOW communication with ground-side software. **[REVISED]** Through DroneBridge (mother) and the USB ESP32 gateway (workers), with mavlink-router or MAVProxy sharing every link with Mission Planner.
- Implement mission state tracking and lost-link handling. **[REVISED]** Rely on ArduPilot's ground-station-loss failsafe on the real aircraft; the coordinator declares an agent lost after missed heartbeats and reassigns its unsearched cells.
- **[REVISED]** Develop the local LLM mission-brief translator (operator brief to validated JSON mission spec) and after-action summarizer. The deterministic planner makes every mission decision (Section 5.5). *(Was: Develop local LLM-assisted mission planning.)*
- Simulate multiple autonomous agents. **[REVISED]** Give each simulated agent a camera footprint and a detection oracle with adjustable miss and false-alarm rates.
- Integrate navigation, perception events, and task reassignment.
- **[NEW]** Use pymavlink rather than DroneKit, which is unmaintained and fails on recent Python versions without patches.

Fall Deliverable: Working three-agent swarm mission simulation.

[NEW] Interface between Members 3 and 4: the AI and autonomy software is split across these two roles. Agree on the detection-event format by October 16: vehicle ID, timestamp, bounding box, confidence, estimated ground position, and frame reference.

Shared Team Responsibilities

All four team members will participate in:

- System architecture decisions.
- Weekly engineering meetings.
- Integration testing.
- Risk management.
- Budget tracking.
- Technical documentation.
- Presentation preparation.
- Demonstration and testing.
- **[NEW]** Flight authorization and compliance records.

Recommended Weekly Commitment: Approximately 6–8 hours per person, adjusted based on course requirements and integration workload.

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3. MASTER PROJECT TIMELINE

WEEK 2: OCTOBER 7–9, 2026

Phase 1 — Project Definition and Research

Primary Goal: Define exactly what the system must accomplish before committing money to hardware.

Tasks

- Conduct a team kickoff meeting.
- Assign primary engineering responsibilities.
- Confirm project expectations with the faculty advisor.
- Clarify whether all three drones must fly simultaneously for the final demonstration.
- Define minimum viable project requirements.
- Write the project problem statement.
- Define measurable engineering success criteria.
- Draw a preliminary overall system block diagram.
- Identify required sensors, processors, motors, and communication interfaces.
- Research verified ArduPilot-compatible flight controllers.
- Compare worker drone propulsion and flight-controller options.
- Estimate worker drone weight, motor thrust, and battery requirements.
- Research battery, GPS, camera, and communication hardware.
- Contact UCI FABWorks and other facilities regarding fabrication availability.
- Investigate loaner flight controllers, radios, and electronics.
- Create a GitHub repository and shared documentation folder.
- Create a hardware budget spreadsheet.
- Install required development software.

Software Setup

- Python
- Visual Studio Code
- Git and GitHub
- ArduPilot SITL
- Mission Planner or a compatible ground control station
- OpenCV
- Ultralytics YOLOv8
- ESP32 development tools
- **[NEW]** pymavlink, and mavlink-router or MAVProxy
- **[NEW]** DroneBridge for ESP32 (flashed with its web flasher)
- **[NEW]** Ollama or llama.cpp for the local LLM
- **[NEW]** An image labeling tool for the detection dataset

Deliverables by Friday, October 9

1. Written project description.
2. Preliminary architecture diagram.
3. Team responsibility assignments.
4. Initial hardware research and estimated BOM.

5. Development environment established.
6. List of technical risks and unresolved requirements.

[NEW] Any Week 2 item still open on October 10 carries into Week 3.

WEEK 3: OCTOBER 12–16, 2026

Phase 2 — Final Design and Hardware Selection

Primary Goal: Finalize the aircraft architectures and prepare an approved procurement list.

Mother Drone Tasks

- Choose a verified ArduPilot-compatible FC/ESC stack. **[REVISED]** Confirm the exact DolphinRC F405 V3 has an ArduPilot target with matching gyro and barometer chips. If that is not confirmed by October 16, switch to the SpeedyBee F405 V3 50A stack.
- Select a compatible motor and propeller combination.
- Determine required battery voltage and capacity.
- Choose GPS with compass.
- Select camera and ESP32 interface. **[REVISED]** ESP32-CAM facing straight down; DroneBridge ESP32 on a spare flight-controller UART for MAVLink.
- Confirm frame dimensions and mounting patterns.
- Create wiring and power-distribution diagrams.
- Verify receiver and failsafe requirements. **[REVISED]** Identify the source of the RC transmitter and receiver (borrow first).

Worker Drone Tasks

- Complete a preliminary mass budget.
- Calculate estimated thrust-to-weight ratio.
- Compare brushed and brushless motor architectures.
- Choose an IMU and embedded flight controller.
- Select motor drivers and battery regulation.
- Choose camera and GPS modules.
- Determine whether the selected propulsion can carry the entire payload.
- Identify which parts must be prototyped before duplicate purchases.
- **[NEW]** Evaluate ESP-Drone firmware on the ESP32-WROOM-32D and record whether it builds and runs.
- **[NEW]** Present the workers' sensing limits (no barometer or compass) at the design review.

Software Tasks

- Successfully run ArduPilot SITL. **[REVISED]** Three instances, each with a unique SYSID_THISMAV.
- Simulate an autonomous waypoint mission.

- Run YOLOv8 on sample search footage. **[REVISED]** Use overhead footage taken from 10–20 m.
- **[REVISED]** Adopt MAVLink v2, list the messages each vehicle sends, and define the detection-event format. *(Was: Define communication message structure.)*
- Define unique drone IDs. **[REVISED]** As MAVLink system IDs.
- Develop the initial search-area allocation algorithm. **[REVISED]** Coverage grid of about 5 m cells, with area split by speed × endurance.
- **[NEW]** Define the search target, number of targets, search-area size, and flight altitude (10–20 m above ground). A person lying down is recommended, since it is the case that justifies a learned detector.
- **[NEW]** Buy one ESP32-CAM and a USB programmer (about $10) and start collecting overhead images from a pole or parking structure at 5–15 m.

Procurement Tasks

- Build a final itemized BOM.
- Identify vendors and exact part variants.
- Verify component compatibility.
- Obtain final prices including tax and shipping.
- Estimate delivery dates.
- Confirm available UCI equipment.
- Obtain advisor approval before purchasing.
- **[NEW]** Itemize Section C (radio, programming, shared hardware, and compliance).
- **[NEW]** Compute delivered cost per vendor (item price, about 7.75% sales tax, shipping), and check each vendor's ship-from location and delivery date.

Flight Authorization Tasks [NEW]

- Each pilot passes the FAA TRUST test.
- Register pilots and the mother drone in the UC Drone Web App and submit the first flight request to UCI EH&S (questions: dronesafety@uci.edu). Moved from Week 7.
- Identify a test site away from people and buildings, and record its dimensions.
- Read the provisions on autonomous flight in UCI's drone procedure before scheduling autonomous tests.
- Decide the mother's Remote ID approach: broadcast module, or flights only at an FAA-Recognized Identification Area.
- Ask EH&S in writing whether simultaneous multi-drone flight can be approved, and under what conditions.

Advisor Tasks [NEW]

- Get a written decision on whether the baseline demonstration (real mother plus simulated agents) satisfies the capstone.
- Map this plan's milestones to the actual EECS 159A graded deliverables and due dates.

Deliverables by Friday, October 16

1. Final hardware architecture.

2. Approved procurement list.
3. Complete system and electrical block diagrams.
4. Mother drone component compatibility review.
5. Worker drone mass/thrust feasibility analysis.
6. Successful initial software simulation.
7. **[NEW]** Communications architecture (Section 5.3).
8. **[NEW]** Flight request submitted and test site identified.
9. **[NEW]** Target, search-area, and altitude definition.
10. **[NEW]** Revised performance targets (Section 6.4).
11. **[NEW]** Advisor decision on the baseline demonstration.

MILESTONE 1: DESIGN APPROVAL

No major purchases should occur until component compatibility, worker propulsion feasibility, and the total delivered cost have been reviewed. **[REVISED]** Also required: a confirmed ArduPilot target for the flight controller being purchased.

WEEK 4: OCTOBER 19–23, 2026

Phase 3 — Hardware Procurement and Mechanical Design

Primary Goal: Order critical components and begin detailed hardware and software development.

Hardware Procurement

- Purchase approved mother flight controller and ESC.
- Purchase mother motors and propellers. **[REVISED]** Include two spare propeller sets.
- Purchase GPS/compass module.
- Purchase required ESP32 modules. **[REVISED]** Include the two DroneBridge boards (air and ground) and the worker ESP-NOW gateway board.
- Purchase the first worker propulsion prototype.
- Purchase first worker IMU and power electronics.
- Purchase required batteries from a suitable supplier.
- Obtain or borrow compatible emergency-control equipment. **[REVISED]** RC transmitter and receiver for the mother.
- Verify and record all order confirmations.
- Record actual costs and estimated delivery dates.

Mechanical Engineering

- Design the mother frame in CAD.
- Design the first worker frame.
- Design camera mounting brackets. **[REVISED]** The mother's camera faces straight down.
- Design GPS mounting locations.
- Design battery restraints.

- Design electronics compartments.
- Prepare initial 3D printing files.

Electrical Engineering

- Produce mother wiring diagram.
- Produce worker motor-driver circuit.
- Identify required voltage regulators.
- Specify UART, I2C, PWM, and GPIO connections.
- Verify expected motor current and ESC capacity.
- Plan grounding, wiring protection, and battery monitoring.
- **[NEW]** Assign a spare flight-controller UART to the DroneBridge ESP32.

Software Engineering

- **[REVISED]** Set up pymavlink with mavlink-router or MAVProxy so Mission Planner and the coordinator share each link; no custom parser. *(Was: Implement basic ground-station telemetry parsing.)*
- Implement the simulated search grid. **[REVISED]** Coverage grid with searched and unsearched cells.
- Create basic agent assignment logic. **[REVISED]** Area split by speed × endurance.
- Create a preliminary detection interface. **[REVISED]** Using the agreed detection-event format.
- Set up logging for telemetry and mission events.
- **[NEW]** Implement the vehicle-adapter interface and the virtual agent (camera footprint plus detection oracle).
- **[NEW]** Measure the ESP32-CAM Wi-Fi stream's frame rate, delay, and percentage of frames delivered at 10, 25, and 50 m. Moved earlier from Week 7.
- **[NEW]** Label the first batch of overhead images.

Deliverables by Friday, October 23

1. Hardware orders placed.
2. Updated budget ledger.
3. Mother and worker CAD designs.
4. Wiring schematics.
5. Working simulated search-area assignment.
6. Initial software repository structure.
7. **[NEW]** Video range measurements.
8. **[NEW]** Vehicle-adapter interface running with at least one SITL agent.

MILESTONE 2: PROCUREMENT COMPLETE FOR THE FIRST PROTOTYPES

WEEK 5: OCTOBER 26–30, 2026

Phase 4 — Fabrication and Initial Hardware Testing

Primary Goal: Begin physical construction and determine whether the worker drone is mechanically feasible.

Mother Drone Tasks

- Print first frame prototype.
- Inspect dimensions and motor mounting points.
- Verify structural integrity.
- Test flight controller power input.
- Check ESC connectivity.
- Verify GPS and compass connections.
- Prepare assembly layout.

Worker Drone Tasks

- Print initial micro-drone frame.
- Weigh all electronics.
- Measure total projected flying mass.
- Build a safe motor-thrust test fixture.
- Measure motor thrust and current draw.
- Evaluate expected hover throttle and battery load.
- Check motor and propeller clearance.
- Decide whether the propulsion configuration is viable.
- **[NEW]** Compare measured mass and thrust with the estimate (about 90–110 g against 120–180 g of thrust).

Software Tasks

- **[REVISED]** Demonstrate the ESP32-CAM stream to the ground station (camera bought in Week 3). *(Was: Demonstrate first ESP32 camera stream if parts have arrived.)*
- Run YOLOv8 on the stream or substitute footage. **[REVISED]** Record a pretrained baseline on the team's overhead images, then start fine-tuning on aerial data.
- Simulate three drones operating in separate search regions. **[REVISED]** With camera footprints and coverage tracking.
- Display basic mission progress. **[REVISED]** Coverage percentage over time.
- Log telemetry messages and timestamps.

Deliverables by Friday, October 30

1. Printed initial aircraft frames.
2. Worker thrust test results.
3. Updated worker mass budget.
4. Initial hardware electrical tests.
5. Functional AI image-processing pipeline.
6. Three-drone search simulation.

MILESTONE 3: WORKER PROPULSION FEASIBILITY GATE

If the proposed worker motors cannot safely lift the required hardware with adequate control margin, redesign the worker propulsion system before buying parts for Worker 2. **[REVISED]** If Worker 1 fails the gate, the Worker 2 reserve ($33.13 plus tax) can be released to the mother's spares and compliance items (Section 4.1).

WEEK 6: NOVEMBER 2–6, 2026

Phase 5 — Electronics Assembly and Embedded Systems Integration

Primary Goal: Assemble the mother drone and worker electronics while establishing basic communication.

Mother Drone Tasks

- Install motors and ESC.
- Mount the flight controller.
- Install GPS/compass.
- Install camera and ESP32 module. **[REVISED]** ESP32-CAM facing down; DroneBridge ESP32 on a spare UART.
- Install radio receiver.
- Complete wiring.
- Verify voltage levels and polarity.
- Connect the FC to ground-control software. **[REVISED]** Over DroneBridge as well as USB.
- Verify sensor readings.
- Verify motor identification and direction with propellers removed.

Worker Drone Tasks

- Assemble worker motor-driver circuitry.
- Connect IMU to ESP32.
- Read and calibrate IMU measurements.
- Connect and test motor control outputs.
- Configure battery voltage monitoring.
- Verify regulated power to electronics.
- Connect GPS receiver.
- Implement basic motor mixing.
- Implement motor-disarm behavior.
- Test battery-current and motor-temperature limits.

Communications Tasks

- **[REVISED]** Bring up the DroneBridge air and ground units (mother MAVLink over

ESP-NOW) and the worker ESP-NOW to USB gateway. *(Was: Establish ESP-NOW connection between development boards.)*
- Transmit unique vehicle identifiers. **[REVISED]** As MAVLink system IDs.
- Transmit heartbeat packets. **[REVISED]** MAVLink HEARTBEAT.
- Transmit sensor telemetry. **[REVISED]** MAVLink GLOBAL_POSITION_INT and SYS_STATUS.
- Add message timestamps.
- Measure packet delivery. **[REVISED]** At least 1,000 messages at each distance, with video running.
- Detect communication loss.
- Define a safe response to link failure. **[REVISED]** ArduPilot ground-station-loss failsafe on the mother; the coordinator declares the agent lost and reassigns its cells.

Software Tasks

- Display telemetry in the ground station.
- Run YOLOv8 on available camera feeds.
- Build a basic mission-state interface.
- Simulate battery warnings and vehicle disconnection.

Deliverables by Friday, November 6

1. Assembled mother drone electronics.
2. Worker motor and IMU bench prototype.
3. Verified ESP-NOW communication. **[REVISED]** MAVLink over ESP-NOW, with loss and delay measured.
4. Working telemetry interface.
5. Electrical and safety test records.

MILESTONE 4: ELECTRICAL INTEGRATION COMPLETE

WEEK 7: NOVEMBER 9–13, 2026

Phase 6 — Initial Flight Validation and System Testing

Primary Goal: Begin controlled flight validation where approved and integrate real hardware telemetry.

Mother Drone Tasks

- Complete sensor calibration.
- Verify battery and voltage monitoring.
- Configure flight modes.
- Test radio disarm and failsafe functions.
- Complete flight-readiness inspection.
- **[REVISED]** Confirm the approved flight request covers this test (request submitted in

Week 3). *(Was: Obtain required testing authorization.)*
- Perform initial supervised hover testing if ready.
- Record flight logs and video.
- Identify stability, vibration, and power issues.
- **[NEW]** Configure and test the polygon geofence and ground-station-loss failsafe (in SITL first).
- **[NEW]** File a post-flight report in the UC Drone Web App after each flight.

Worker Drone Tasks

- Verify IMU axis orientation.
- Verify motor mixing.
- Test stabilization outputs.
- Evaluate flight-control loop timing.
- Validate the loss-of-link disarm function.
- Conduct staged testing only when propulsion and safety requirements are met.
- Record motor-response and stability measurements.

Ground Station Tasks

- Receive mother drone telemetry.
- Display location and operating state.
- Record camera stream frame rate. **[REVISED]** On the aircraft; the bench range test was done in Week 4.
- Evaluate camera transmission latency.
- Demonstrate YOLOv8 detections.
- Log detection events.

Swarm Tasks

- Simulate three vehicles executing separate search routes.
- Demonstrate mission reassignment.
- Test disconnected vehicle handling.
- Integrate simulated mission states with ground-station telemetry. **[REVISED]** Run the mixed configuration: the real mother (on the bench, or in flight if authorized) plus two SITL agents in one coordinator.

Deliverables by Friday, November 13

1. Mother drone initial flight results or documented blocker.
2. Worker stabilization test results.
3. Ground station displaying real telemetry.
4. Camera transmission performance measurements.
5. Integrated simulation demonstration. **[REVISED]** Including the real mother's telemetry.

MILESTONE 5: INITIAL SYSTEM VALIDATION

WEEK 8: NOVEMBER 16–20, 2026

Phase 7 — Prototype Refinement and Design Freeze

Primary Goal: Resolve major hardware problems and lock the architecture for Winter integration.

Mother Drone Tasks

- Fix issues discovered during initial testing.
- Refine ArduPilot parameters.
- Verify GPS position estimates.
- Validate emergency-control procedures.
- Test waypoint mission behavior in simulation.
- Conduct additional authorized flight tests if appropriate.

Worker Drone Tasks

- Improve stabilization and motor responsiveness.
- Reduce unnecessary airframe weight.
- Improve electronic mounting.
- Resolve power instability.
- Evaluate whether a custom PCB would improve the system.
- Finalize the worker design.
- Approve second worker purchase only after prototype validation.

Software Tasks

- Integrate live detections into the mission coordinator.
- Add mission event logging.
- Implement task reassignment when an agent becomes unavailable. **[REVISED]** Its unsearched cells go to the remaining agents.
- Test simulated collision avoidance and no-fly boundaries.
- Develop a more complete ground station interface.

Documentation Tasks

- Update complete CAD designs.
- Update wiring schematics.
- Record firmware and software versions.
- Update BOM with actual costs.
- Update risk assessment.
- Conduct advisor review.
- **[NEW]** Update flight authorization records.

Deliverables by Friday, November 20

1. Validated mother design.
2. Finalized worker architecture.
3. Updated software architecture.
4. Approved second worker build strategy.
5. Updated test evidence and risk register.

MILESTONE 6: HARDWARE ARCHITECTURE FREEZE

After this deadline, avoid major design changes unless required for safety or fundamental functionality.

WEEK 9: NOVEMBER 23–27, 2026

Phase 8 — System Demonstration and Report Development

Primary Goal: Produce a repeatable prototype demonstration and draft the Fall technical report.

Thanksgiving holidays: November 26–27. Complete important team testing before the holiday.

Hardware Tasks

- Repair outstanding mother drone issues.
- Verify repeatable system startup.
- Check all electrical connections.
- Verify camera and telemetry operation.
- Continue Worker 1 testing.
- Assemble Worker 2 if validated parts are available.
- Document the current physical configuration of each aircraft.

Software Tasks

- Run YOLOv8 detection using an actual ESP32 camera feed.
- Demonstrate telemetry transmission to the laptop.
- Run three-agent simulated search.
- Demonstrate automatic region assignment.
- Demonstrate simulated lost-agent reassignment.
- Record end-to-end event logs.
- **[NEW]** Geotag detections and measure location error against surveyed target positions.

Documentation Tasks

- Draft technical report.
- Update project architecture diagrams.
- Collect photographs of hardware.
- Export CAD files.
- Include electrical schematics.

- Include BOM and actual expenses.
- Summarize completed tests.
- Identify remaining Winter tasks.
- Begin presentation slides.

Deliverables by Wednesday, November 25

1. Repeatable preliminary demonstration.
2. Draft Fall technical report.
3. Current hardware and software documentation.
4. Preliminary presentation.
5. List of unfinished requirements.

MILESTONE 7: DEMONSTRATION FREEZE

WEEK 10: NOVEMBER 30 – DECEMBER 4, 2026

Phase 9 — Final Verification and Fall Project Completion

Primary Goal: Complete engineering work and prepare the final Fall submission.

Testing Tasks

- Perform final mother hardware bench checks.
- Confirm stable telemetry and camera streaming.
- Re-run approved flight tests where appropriate.
- Verify worker sensor and motor functions.
- Validate emergency shutdown behavior.
- Run final integrated ground station demonstration.
- Demonstrate three-agent mission allocation in simulation.
- Record successful and unsuccessful test results.
- Measure performance metrics. **[REVISED]** Against the targets in Section 6.4.

Documentation Tasks

- Finalize technical report.
- Finalize bill of materials.
- Finalize CAD drawings.
- Finalize wiring diagrams.
- Complete design justification.
- Complete risk analysis.
- Summarize engineering calculations.
- Document testing procedures and outcomes.
- Finalize Winter Quarter execution plan.

Presentation Tasks

- Prepare final presentation slides.
- Assign speaking sections.
- Record demonstration footage.
- Rehearse the full presentation.
- Obtain advisor feedback.
- Make final corrections.

Deliverables by Friday, December 4

1. Completed Fall engineering prototype package.
2. Final draft technical report.
3. Finished demonstration materials.
4. Completed presentation.
5. Detailed Winter project plan.

MILESTONE 8: FALL ENGINEERING WORK COMPLETE

FINALS WEEK: DECEMBER 5–11, 2026

Phase 10 — Submission and Winter Handoff

Primary Goal: Submit all required materials and prepare the team to resume in Winter.

Tasks

- Submit required EECS 159A assignments by their actual Canvas deadlines.
- Deliver project presentation and demonstration.
- Submit final technical report.
- Back up GitHub repository and project files.
- Archive hardware firmware versions.
- Archive CAD and PCB files.
- Archive test data and flight logs.
- Save purchase receipts and budget records.
- Document remaining problems.
- Assign Winter Week 1 tasks.
- Properly disconnect and store LiPo batteries.
- Confirm that all teammates have access to project files.

Deliverables by December 11

1. Complete Fall report and presentation.
2. Functional preliminary prototype and recorded test evidence.
3. Updated project repository.

4. Complete engineering documentation.
5. Winter Quarter development roadmap.

MILESTONE 9: FALL QUARTER COMPLETE

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4. HARDWARE PROCUREMENT AND BUDGET MANAGEMENT

4.1 Budget Distribution

Maximum Total Hardware Budget: $300

[REVISED] The table now shows the original and revised allocations side by side.

Category	Original planning budget	Revised planning budget
Mother drone	$121.61	$149.94
Worker drones (2 total)	$66.26	$66.26, with Worker 2's $33.13 held until the October 30 gate
Radio, programming, shared hardware, and compliance (Section C)	$70.86, not itemized	$18.90 itemized; RC link and Remote ID paid from the remainder
Estimated sales tax (about 7.75% on goods)	Included in the line below	$17.83
Remaining for shipping, RC link, Remote ID, and contingency *(Was: Remaining taxes, shipping, and contingency)*	$41.27	$47.07
Total	**$300.00**	**$300.00**

[REVISED] The known-item subtotal is $235.10 before tax and $252.93 with estimated tax. It is a preliminary sourcing estimate, not a verified final checkout amount. All taxes, delivery charges, and replacement components must remain within the $300 limit. *(Was: "The $258.73 component subtotal is a preliminary sourcing estimate...")*

Open budget decisions [NEW]

The $47.07 remainder must cover shipping, the RC link, and Remote ID. If it falls short, apply these in order:

1. Consolidate orders to as few vendors as possible. At an assumed $5 per order, nine

separate orders would use most of the remainder.
2. Borrow the RC transmitter and receiver, a second 3S battery, and a USB programmer through UCI or a campus club.
3. For Remote ID, fly the mother only at an FAA-Recognized Identification Area or borrow a module, rather than buying one.
4. If the DolphinRC stack's ArduPilot target is confirmed, buy it instead of the SpeedyBee (returns $16.59 including tax).
5. Defer the Worker 2 purchase and represent Worker 2 in simulation (returns $35.70 including tax).
6. Request additional funds from the advisor or department.

4.2 Purchasing Schedule

October 7–16

- Complete hardware selection.
- Confirm component compatibility.
- Check UCI inventory and equipment availability.
- Compare vendors.
- Obtain final delivered-price estimates.
- Obtain design approval.
- **[NEW]** Buy one ESP32-CAM and a USB programmer for vision dataset work (about $10; the camera is the mother's).

October 19–23

- Purchase critical mother drone components.
- Purchase one worker prototype's essential components.
- Order communication electronics and required sensors. **[REVISED]** Including the three ESP32 boards for DroneBridge and the worker gateway.
- Place battery orders after compatibility checks.

October 26–30

- Receive and inspect early components.
- Complete initial propulsion testing.
- Resolve any component or motor incompatibility.

November 2–20

- Purchase verified replacement parts if needed.
- Purchase Worker 2 components only after validating Worker 1. **[REVISED]** If Worker 1 fails the gate, release the Worker 2 reserve as described in Section 4.1.
- Reserve sufficient funds for tax, shipping, and unforeseen costs.

4.3 UCI Resources

The team should attempt to source the following through UCI facilities:

- 3D-printed airframes.

- Motor mounts.
- Camera mounts.
- GPS mounts.
- Propeller guards.
- Landing gear.
- Electronics enclosures.
- Battery holders.
- Prototype PCB materials.
- Soldering stations.
- Solder, flux, wire, and heat shrink.
- Multimeters.
- Oscilloscopes.
- Bench power supplies.
- LiPo balance chargers and safety equipment.
- General machining and fabrication tools.
- **[NEW]** RC transmitter and receiver (loan).
- **[NEW]** Spare 3S LiPo batteries (loan).
- **[NEW]** USB-UART programmer.

Relevant campus and external resources:

- UCI FABWorks: https://manufacturing.uci.edu/index.php/fabworks-2/
- UCI IDMI Machine Shop: https://manufacturing.uci.edu/index.php/facilities/machineshop/
- UCI EECS Capstone: https://sites.uci.edu/eecscapstone/
- JLCPCB: https://jlcpcb.com/
- PCBWay: https://www.pcbway.com/
- McMaster-Carr: https://www.mcmaster.com/
- **[NEW]** UCI EH&S Drone Safety: https://ehs.uci.edu/drone-safety

Free access or materials should be confirmed with the facility; do not assume all services are automatically covered.

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5. SOFTWARE DEVELOPMENT PLAN

5.1 Software Architecture

Ground Laptop

- YOLOv8 for object detection.
- OpenCV for image processing.
- Python for telemetry and mission logic.
- Local LLM for high-level mission assistance. **[REVISED]** Limited to mission-brief translation and after-action summaries (Section 5.5).
- Ground station dashboard.
- Mission coordination system.
- Search-area allocation algorithm.

- **[NEW]** MAVLink router (mavlink-router or MAVProxy) and pymavlink.
- **[NEW]** Vehicle-adapter layer (MAVLink, worker, virtual agent).
- **[NEW]** Coverage grid and capability-weighted allocator.
- **[NEW]** Detection geotagging.

Mother Drone

- ArduPilot flight-control firmware.
- GPS and compass.
- ESP32 camera interface. **[REVISED]** ESP32-CAM facing straight down, streaming MJPEG over Wi-Fi to the team access point.
- Telemetry connection. **[REVISED]** MAVLink through DroneBridge for ESP32 over ESP-NOW to a USB ground unit.
- Receiver-based safety control.
- **[NEW]** ArduPilot polygon geofence and ground-station-loss failsafe.

Worker Drones

- ESP32 embedded firmware. **[REVISED]** Evaluate ESP-Drone as the base.
- IMU sensor processing.
- Motor control and stabilization.
- GPS telemetry. **[REVISED]** Sent as MAVLink over ESP-NOW.
- ESP32 camera streaming.
- Communication loss monitoring.
- Emergency disarm system.

5.2 Software Implementation Order

1. Establish development environment.
2. Set up repository and version control.
3. Run ArduPilot simulation. **[REVISED]** Three instances with unique system IDs.
4. Implement simulated waypoint navigation.
5. Install and test YOLOv8.
6. Implement image detection on prerecorded footage. **[REVISED]** Overhead footage from the team's ESP32-CAM at 10–20 m.
7. Develop simulated search-area allocation. **[REVISED]** Coverage grid, with area split by speed × endurance.
8. **[REVISED]** Adopt MAVLink v2 and list each vehicle's messages. *(Was: Define drone telemetry messages.)*
9. Develop ESP32 Wi-Fi communications. **[REVISED]** Video only.
10. Develop ESP-NOW telemetry and control messages. **[REVISED]** DroneBridge for the mother; ESP-NOW to USB gateway for the workers.
11. Implement camera streaming.
12. Integrate live telemetry into the laptop. **[REVISED]** Through mavlink-router or MAVProxy and pymavlink.
13. Build the ground station dashboard. **[REVISED]** Use cached map tiles or local coordinates; the field laptop has no internet.
14. Integrate detection events with mission coordinator.
15. Implement mission reassignment. **[REVISED]** Reassign unsearched cells and upload new

AUTO missions.
16. Simulate connection failures and low-battery behavior.
17. Conduct end-to-end hardware/software integration. **[REVISED]** Real mother plus simulated agents first.
18. Record benchmarks and test results.
19. **[NEW]** Implement the vehicle-adapter interface and virtual agents (do this right after step 4).
20. **[NEW]** Geotag detections and measure location error.

The local LLM should provide high-level assistance, not unrestricted real-time motor control. Each aircraft requires deterministic stabilization and safety behavior. **[REVISED]** The deterministic planner makes every mission decision (Section 5.5).

5.3 Communications Architecture [NEW]

- **Topology:** star. Every vehicle talks directly to the ground station. The mother does not relay worker traffic; relay is future work. All flights stay within visual line of sight, where each vehicle can reach the ground directly.
- **Message format:** MAVLink v2 for every vehicle and every SITL instance, with a unique system ID per vehicle.
- **Mother link:** an ESP32 running DroneBridge for ESP32 sits on a spare flight-controller UART and sends MAVLink over ESP-NOW (long-range mode) to a second ESP32 on the laptop's USB port. DroneBridge is telemetry-only.
- **Worker link:** the worker flight controller sends MAVLink HEARTBEAT, GLOBAL_POSITION_INT, and SYS_STATUS over ESP-NOW (each fits in one ESP-NOW packet) to an ESP32 gateway on the laptop's USB port. A laptop's Wi-Fi cannot send or receive ESP-NOW.
- **Sharing links:** mavlink-router or MAVProxy gives Mission Planner and the coordinator access to every vehicle at once.
- **Video:** each ESP32-CAM streams MJPEG over plain Wi-Fi to a team-owned access point (laptop hotspot or travel router), not campus Wi-Fi, whose login system and device isolation block it. One 640×480 stream at 10 fps needs roughly 2–3 Mbit/s, about ten times DroneBridge's ESP-NOW rate of about 250 kbit/s, so video never shares the telemetry link.
- **Range:** the ESP32-CAM's onboard antenna is expected to hold a usable stream at tens of meters, not hundreds. The Week 4 range test sets the search-area size.
- **Safety link:** RC transmitter and receiver on the mother, independent of the autonomy software.

5.4 Vision Pipeline Requirements [NEW]

- **Target:** decide by October 16. A person lying down is recommended, since that is the case that justifies a learned detector. A high-visibility tarp can be found with a color threshold, and reviewers will ask why a neural network was used.
- **Altitude:** 10–20 m above ground. With the stock OV2640 lens at 640×480, the ground footprint is about 1 to 1.3 times the altitude, so a person lying down is about 85–110 pixels long at 10 m and about 30 pixels at 30 m.

- **Model:** start from pretrained YOLOv8 and expect weak results on small, prone people seen from overhead. Fine-tune on aerial datasets (VisDrone, HERIDAL, SARD) plus the team's own images.
- **Dataset:** collect overhead images with the actual ESP32-CAM before the drone flies, from a pole or a parking structure at 5–15 m, with team members lying on grass in varied clothing and poses. Keep a held-out test set from separate sessions.
- **Evaluation:** at least 200 target instances in the held-out test set; report precision and recall at a fixed confidence threshold. With only 20 images, a measured 90% recall has a 95% confidence interval of about 70–97%; with 200 instances it narrows to about ±4 percentage points.
- **Geotagging:** report each detection's ground position from camera orientation (straight down), lens field of view, altitude, attitude, and timestamps. The simplest approach stamps both video frames and telemetry as they reach the laptop, then corrects for a measured delay. Measure location error against surveyed target positions.
- **Compute:** confirm the ground laptop's GPU, RAM, and OS before committing to three streams plus a local LLM.

5.5 Local LLM Role [NEW]

- The deterministic planner (coverage grid, area allocation, reassignment) makes every mission decision.
- The LLM converts an operator's plain-language brief into a JSON mission spec (search polygon, priority zones, altitude). A validator checks the spec against the geofence and battery limits before it is used.
- The LLM writes the after-action summary from the detection and event logs.
- Output is constrained to a JSON schema; llama.cpp and Ollama both support schema-constrained output.
- The system runs the same with the LLM switched off.
- Evaluation: about 30 scripted briefs, reporting the share that produce a valid spec and the share that match the intended mission.

5.6 Swarm Coordinator Design [NEW]

- **Vehicle adapters:** one interface with three implementations: MAVLink (SITL and the real mother), worker, and virtual agent. Adding the real mother is a configuration change, not new code.
- **Virtual agents:** each has a camera footprint and a detection oracle that fires when the footprint covers a planted target, with adjustable miss and false-alarm rates. Gazebo is not needed.
- **Coverage grid:** cells of about 5 m, marked searched as each footprint passes. Coverage percentage comes directly from the grid, and reassignment hands out unsearched cells.
- **Allocation:** area is split by each vehicle's speed × endurance, not into equal thirds.
- **Mission upload:** missions go to ArduPilot as AUTO missions, and reassignment uploads a new mission, so a vehicle keeps searching through a link dropout.
- **Safety:** ArduPilot's polygon geofence and ground-station-loss failsafe are configured on the real aircraft, so a coordinator error cannot send it outside its region.
- **Lost agent:** declared after N missed heartbeats, with N set from measured link behavior;

its unsearched cells are reassigned.
- **Library:** pymavlink. DroneKit is unmaintained and fails on recent Python versions without patches.
- **Field display:** the laptop has no internet at the field, so cache map tiles or display in local coordinates.

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6. HARDWARE AND SYSTEM TESTING REQUIREMENTS

6.1 Mother Drone Tests

- Verify electrical polarity.
- Verify safe power distribution.
- Verify flight-controller startup.
- Verify ESC connectivity.
- Verify motor order and direction.
- Calibrate sensors.
- Validate GPS and compass.
- Verify receiver connectivity.
- Verify emergency disarm.
- Verify communication-loss failsafe.
- Verify battery monitoring.
- Complete flight-readiness checklist.
- Perform authorized supervised hover testing.
- Analyze logs and document stability.
- **[NEW]** Verify the geofence and ground-station-loss failsafe in SITL, then on the aircraft.
- **[NEW]** Verify the DroneBridge telemetry link and measure its range.

6.2 Worker Drone Tests

- Verify electronics mass.
- Verify motor thrust.
- Verify battery current and voltage.
- Validate regulator output.
- Validate IMU data.
- Verify motor mixing.
- Test control-loop timing.
- Test emergency disarm.
- Verify camera stream.
- Verify GPS telemetry.
- Verify ESP-NOW connection. **[REVISED]** MAVLink messages received at the ground gateway.
- Evaluate flight stabilization.

- Validate safe failure behavior.

6.3 Ground Station Tests

- Receive telemetry from a real ESP32.
- Display live drone status.
- Receive camera frames.
- Record video frame rate and latency.
- Run YOLOv8 detection.
- Detect designated mock rescue targets. **[REVISED]** The target defined in Section 5.4.
- Log object-detection confidence.
- Track detection timestamps.
- Simulate three assigned search regions.
- Demonstrate mission reassignment after loss of one simulated agent. **[REVISED]** Measured against Section 6.4.
- Record results and error conditions.
- **[NEW]** Measure detection geolocation error.
- **[NEW]** Run the coordinator with the real mother plus SITL agents.

6.4 Preliminary Performance Targets

[REVISED] The table now shows the original and revised targets side by side.

Metric	Original Fall target	Revised Fall target
Camera test duration	60 seconds continuous streaming	Frame rate, delay, and percentage of frames delivered at 10, 25, and 50 m, at 640×480
ESP-NOW test	100 consecutive messages with packet-loss measurement	At least 1,000 messages at each distance with video running; loss percentage and delay
AI test dataset	At least 20 labeled test images	At least 200 held-out target instances from the team's camera at flight altitude; precision and recall at a fixed threshold
Swarm simulation	Three active simulated agents	Three agents running through the same MAVLink adapter used by the real mother
Search regions	Three distinct assigned areas	Areas split by speed × endurance; coverage percentage tracked on the grid
Failure test	Detect lost simulated agent and reassign its task	Agent declared lost within N seconds of its last heartbeat; all of its unsearched cells reassigned; 100% final coverage
Mother flight	Supervised controlled hover if flight-ready and authorized	Unchanged

Worker flight	Demonstrated stabilization or documented redesign decision	Unchanged
Total hardware expenditure	$300 maximum delivered	Unchanged; tracked as delivered cost including tax and shipping
[NEW] Search outcome	None	Coverage percentage over time, time to first detection, and geolocation error against true target positions
[NEW] Local LLM	None	Share of about 30 scripted briefs that produce a valid and correct mission spec

These are internal engineering targets and may be adjusted after baseline measurements.

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7. RISK MANAGEMENT

Risk	**Impact**	**Mitigation**
Worker motors cannot lift payload	Critical	Measure actual thrust before duplicate purchases. **[REVISED]** Estimated thrust-to-weight is about 1.2–1.8 against a 2:1 target.
Flight controller incompatible with ArduPilot	High	Verify exact board and firmware target before ordering. **[REVISED]** Check that the gyro and barometer chips match the target; fallback is the SpeedyBee F405 V3 50A stack.
ESP32 camera adds excessive weight	High	Optimize payload and frame before flight
Battery insufficient for propulsion	High	Validate current, voltage sag, and expected flight time
Worker stabilization development takes too long	High	Establish early technical gate and simulation fallback. **[REVISED]** Start from ESP-Drone firmware.
Wireless interference	Medium–High	Test throughput, channel configuration, and packet loss. **[REVISED]** Keep video off the telemetry link.
Autonomous navigation instability	High	Start with SITL and supervised individual flight
Hardware shipping delays	High	Order critical components early. **[REVISED]**

		Check each vendor's ship-from location.
Budget overrun	High	Compare total delivered prices before purchase. **[REVISED]** $47.07 remains after known items and tax; Worker 2 funds are held in reserve (Section 4.1).
Component failure	Medium	Preserve funds and test components individually. **[REVISED]** Two spare propeller sets budgeted for the mother.
Campus fabrication unavailable	Medium	Confirm authorization and scheduling early
Unsafe simultaneous flight	Critical	Require individual flight validation and authorization. **[REVISED]** Under Part 107, one person flying several drones needs a §107.35 waiver; under the recreational rules, get EH&S's answer in writing. One pilot with a kill switch per airborne aircraft.
[NEW] Workers cannot fly autonomous waypoints (no barometer or compass)	Critical	Baseline demonstration uses the real mother plus simulated agents; workers demonstrate stabilization, sensing, camera, and telemetry
[NEW] Flight authorization delayed or denied (EH&S approval, Class C airspace, Remote ID)	Critical	Submit in Week 3; identify an approved site early; decide the Remote ID approach by October 16
[NEW] Video range limits the search area	High	Measure in Week 4 and size the search area to the measured range
[NEW] Pretrained detector misses prone people seen from overhead	High	Fine-tune on aerial data and team images; fly at 10–20 m
[NEW] LLM produces an invalid or unsafe mission spec	Medium	Schema-constrained output and a validator; the system runs with the LLM off
[NEW] Detection locations wrong because video and telemetry timestamps differ	Medium	Timestamp both streams at the laptop, then measure and correct the delay

Contingency Strategy

[REVISED] This configuration is now the baseline plan rather than the fallback (Section 1.1). Autonomous flight by the physical workers becomes a stretch goal.

If all three drones cannot achieve reliable autonomous flight within the budget, the team should preserve a technically meaningful final demonstration consisting of:

1. One operational ArduPilot mother drone.

2. Two physical worker prototypes with verified sensors, communications, and documented flight-control progress.
3. A functioning laptop-based AI vision system.
4. A three-agent autonomous swarm simulation. **[REVISED]** Running through the same MAVLink adapter as the real mother.
5. A working interface between actual hardware telemetry and the simulated mission coordinator.

This fallback should be confirmed with the faculty advisor rather than assumed to satisfy all capstone requirements. **[REVISED]** Get the decision in writing by October 16.

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8. FALL QUARTER TECHNICAL DOCUMENTATION

The team should maintain the following throughout the quarter.

Required Engineering Documents

1. Project proposal and problem statement.
2. Functional and nonfunctional requirements.
3. System architecture diagram.
4. Hardware block diagram.
5. Communications and data-flow diagram.
6. Mother drone wiring diagram.
7. Worker drone wiring diagram.
8. CAD models and manufacturing drawings.
9. Itemized bill of materials.
10. Budget and purchasing records.
11. Power and mass budgets.
12. Motor thrust evaluation.
13. Software architecture documentation.
14. Communication protocol specification. **[REVISED]** Communications architecture and MAVLink message set, with the system ID map.
15. Flight-readiness and safety procedures.
16. Subsystem test procedures.
17. Test results and analysis.
18. Risk register and mitigation plan.
19. Design decisions and revision history.
20. Winter Quarter implementation schedule.
21. **[NEW]** Flight authorization records: UC Drone Web App approvals and post-flight reports, TRUST certificates, FAA registration, and Remote ID compliance.
22. **[NEW]** Dataset description and detector evaluation report.
23. **[NEW]** Detection-event format agreed between Members 3 and 4.

Suggested GitHub Repository Structure

[REVISED] Added folders are labeled NEW.

```
drone-swarm/
├── docs/
├── hardware/
│   ├── mother/
│   ├── worker/
│   └── pcb/
├── firmware/
│   ├── mother-bridge/      (DroneBridge configuration)
│   ├── ground-gateway/     (NEW: worker ESP-NOW to USB gateway)
│   └── worker/
├── ground-station/
│   ├── vision/
│   │   └── dataset/        (NEW)
│   ├── telemetry/
│   └── dashboard/
├── swarm/
│   ├── adapters/           (NEW: MAVLink, worker, virtual agent)
│   ├── planning/
│   ├── agents/
│   ├── llm/                (NEW: brief translator and validator)
│   └── safety/
├── simulation/
└── tests/
```

All design files, software, diagrams, test logs, and documentation should be stored in the shared repository or linked project folder.

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9. WEEKLY PROJECT MANAGEMENT PROCESS

Weekly Team Meeting

Recommended Frequency: Once per week, approximately 30 minutes.

Meeting Agenda

1. Review last week's goals.
2. Confirm completed tasks.
3. Identify incomplete or blocked tasks.

4. Review hardware and software progress.
5. Review budget spending.
6. Review safety and technical risks.
7. Assign next week's tasks.
8. Confirm milestone deadlines.

Weekly Integration Session

Recommended Frequency: Once per week, approximately two hours.

During this session, members should combine their subsystem work, test interfaces, and identify problems that do not appear during isolated development.

Weekly Status Report Format

Week:
Date:
Completed Tasks:
Tasks In Progress:
Blocked Tasks:
Hardware Purchases:
Budget Remaining:
Testing Completed:
Technical Risks:
Next Week's Priorities:
Assigned Owners:
[NEW] Flight Authorization Status:

Every task should have one primary owner and a deadline.

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10. WINTER QUARTER PREVIEW

Fall Quarter is not the final project deadline. The purpose of the Fall plan is to create a sufficiently mature system so Winter can focus on performance and integration.

January 4–15, 2027

Goal: Hardware Completion

- Finalize unresolved hardware.
- Assemble remaining drone components.
- Validate independent electronics and flight controls.
- Verify communications.
- Repair mechanical and electrical failures.
- **[NEW]** Confirm that flight approvals cover the Winter test dates.

January 18–29, 2027

Goal: System Integration

- Integrate all drone telemetry with the ground station.
- Integrate camera video feeds.
- Connect navigation information to the mission coordinator.
- Test target detection and event handling.
- Continue individually authorized flight tests.

February 1–12, 2027

Goal: Swarm Intelligence

- Finalize search-area allocation.
- Implement mission reassignment.
- **[REVISED]** Integrate the local LLM as mission-brief translator and after-action summarizer, with validated outputs. *(Was: Integrate the local LLM for high-level task planning.)*
- Test agent coordination.
- Simulate failures and recovery behavior.

February 15–26, 2027

Goal: Coordinated System Testing

- Verify stable operation of individual aircraft.
- Test multi-agent communications.
- Test search-task coordination.
- Perform supervised cooperative tests only when approved. **[REVISED]** Simultaneous multi-drone flight only with written EH&S approval.
- Measure detection performance and communication reliability.
- Refine mission behavior.

March 1–12, 2027

Goal: Final Demonstration and Validation

- Complete repeatable demonstration.
- Evaluate system reliability.
- Record test results.
- Record demonstration videos.
- Complete final technical documentation.
- Prepare final presentation.
- Correct critical remaining issues.

March 13–19, 2027

Goal: Final Submission

- Submit required final capstone deliverables.
- Deliver final presentation.
- Demonstrate the system.
- Archive source code and documentation.
- Complete the project handoff.

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11. MAJOR PROJECT MILESTONES

Date	Milestone	Expected Result
October 9	Project Definition	Scope, roles, architecture, initial research
October 16	Design Approval	Approved BOM, compatibility, feasibility. **[REVISED]** Plus advisor decision on the baseline, flight request submitted, target and search area defined, and flight controller target confirmed.
October 23	Prototype Procurement	Critical initial hardware ordered. **[REVISED]** Video range measured.
October 30	Worker Feasibility	Motor thrust and weight evaluation. **[REVISED]** Worker 2 reserve decision.
November 6	Electronics Integration	Working sensors, motor control, telemetry. **[REVISED]** MAVLink over ESP-NOW.
November 13	Initial Validation	Mother flight/bench results, worker stabilization. **[REVISED]** Real mother plus SITL agents in one coordinator.
November 20	Architecture Freeze	Verified hardware design and Winter direction
November 25	Demonstration Freeze	Repeatable preliminary system demonstration
December 4	Engineering Completion	Final report draft, testing, presentation
December 11	Fall Quarter Complete	Formal submission and Winter handoff
January 22	Winter Hardware Checkpoint	All required prototypes assembled and tested
February 26	System Integration Target	Coordinated functionality and test evidence

March 12	Final Demonstration Target	Working system and final report ready
March 19	Winter Quarter End	Capstone complete, subject to actual course deadlines

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12. IMMEDIATE ACTION PLAN — OCTOBER 7–9

The following should be completed within the next two days.

- Schedule the first full team meeting.
- Assign the four engineering roles.
- Confirm the faculty advisor and Fall course deliverables.
- Write the final project description.
- Establish the minimum viable final demonstration.
- Create GitHub repository and shared folder.
- Create the first system architecture diagram.
- Verify mother flight-controller compatibility.
- Complete the initial worker propulsion weight estimate.
- Contact UCI fabrication facilities.
- Investigate university equipment loans.
- Create the $300 budget tracker.
- Install the software development environment.
- Run the first simulated ArduPilot mission.
- Schedule the October 16 hardware design review.

12.1 Revised Action Plan, October 12–16 [NEW]

- Advisor: get a written decision on the baseline demonstration, and map milestones to the EECS 159A graded deliverables.
- Confirm the DolphinRC ArduPilot target and chips, or switch to the SpeedyBee F405 V3 stack.
- Complete TRUST certificates, register in the UC Drone Web App, submit the first flight request, identify the test site, and decide the Remote ID approach.
- Freeze the communications architecture (Section 5.3).
- Define the target, search area, and altitude; buy one ESP32-CAM and a programmer; start the overhead image dataset.
- Itemize Section C and compute delivered cost with tax and shipping.
- Adopt the revised performance targets (Section 6.4).
- Confirm the roster and assign the worker lead.
- Agree on the detection-event format between Members 3 and 4.
- Run three SITL instances through the vehicle-adapter interface.

13. FALL QUARTER DEFINITION OF SUCCESS

By December 11, 2026, the team should have completed the fundamental engineering design and have evidence that its major subsystems are operational.

Minimum Successful Fall Outcome

- One assembled ArduPilot mother drone with verified electrical and navigation systems.
- One worker drone prototype with tested propulsion, sensors, electronics, and communications.
- A working ESP32 camera interface.
- A ground laptop running AI detection. **[REVISED]** Evaluated on the held-out overhead test set.
- A simulated three-drone autonomous search mission. **[REVISED]** Running through the same MAVLink adapter as the real mother.
- Verified communication between physical hardware and ground software.
- Complete technical documentation.
- A clear, achievable Winter Quarter roadmap.
- **[NEW]** Flight authorization in place before any outdoor flight.
- **[NEW]** Advisor's written approval of the baseline demonstration.

Stretch Goals

- Successful supervised mother drone autonomous waypoint flight.
- Successful stabilized flight of Worker 1.
- Completed assembly of Worker 2.
- Simultaneous telemetry from multiple physical drone prototypes.
- Real camera detections triggering mission-state updates.
- Early coordinated behavior between physical and simulated agents. **[REVISED]** Promoted to the Winter baseline.
- **[NEW]** Detections geotagged with measured location error.

The team should prioritize reliable engineering, documented verification, and a functioning integrated prototype over attempting unvalidated simultaneous autonomous flight too early.

Final Objective: Enter Winter Quarter with the aircraft architecture established, expensive engineering uncertainties resolved, and the main AI and mission-coordination components already operating. This gives the team the best opportunity to complete a credible autonomous search-and-rescue drone swarm by March 2027 within the $300 hardware limit.

14. REFERENCES [NEW]

- UCI EH&S Drone Safety: https://ehs.uci.edu/drone-safety
- UCI Small UAS/Drone Program procedure (PDF): https://ehs.uci.edu/programs/_pdf/fire-safety/unmanned-aircraft-systems.pdf
- UCI faculty, staff, and student UAS flight request instructions (PDF): https://www.ehs.uci.edu/drone-safety/_pdf/faculty-staff-and-student-uas-flight-request-instructions.pdf
- UCOP guidance on student use of drones in classes (PDF): https://www.ucop.edu/safety-and-loss-prevention/_files/uas/student-use-of-drones-in-classes.pdf
- ArduPilot documentation, SpeedyBee F405 V3: https://ardupilot.org/copter/docs/common-speedybeef4-v3.html
- ArduPilot forum, SpeedyBee F405 V3 porting thread: https://discuss.ardupilot.org/t/approved-speedybee-f405-v3-flight-controller-for-porting/92285
- SpeedyBee F405 V3 50A stack listing (RaceDayQuads): https://www.racedayquads.com/products/speedybee-f405-v3-bls-3-6s-30x30-stack-combo-f405-fc-50a-4in1-esc
- Espressif ESP-Drone (GitHub): https://github.com/espressif/esp-drone
- ESP-Drone documentation: https://docs.espressif.com/projects/espressif-esp-drone/
- DroneBridge for ESP32 (GitHub): https://github.com/DroneBridge/ESP32
- ArduPilot documentation, DroneBridge / ESP32 telemetry: https://ardupilot.org/copter/docs/common-esp32-telemetry.html